Final Project

SUBMARINE MONITORING SYSTEM

Local Node Controller (LNC) End Unit

Software Functional Definition

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1. Introduction

1.1 Purpose

This document defines the software functional behaviour of the Submarine Monitoring System, comprising a Local Node Controller (LNC) end unit, a Central Computer, and a Ground Station. It specifies the responsibilities of each software module, the conditions under which each module acts, and the messages exchanged between them.

1.2 System Context

The LNC is an end unit installed in the submarine and connected via a communication link to the Central Computer, which manages all submarine activity. The LNC hosts environmental sensors — a temperature sensor, a humidity sensor, and a light sensor — monitors the submarine's battery state, and hosts an object-detection component (sonar) that monitors the direction of travel.

The Central Computer manages several end units (LNCs) in addition to this one, such as a motor unit and a navigation unit.

Communication between the Central Computer and the LNC end unit is over UART or Ethernet. The communication layer is designed so that the underlying transport can be replaced with another technology in the future with minimal impact on the rest of the system. Communication between the Central Computer and the Ground Station is over Ethernet, and is used both to receive commands and to transfer data accumulated in the submarine from the various end units.

All the messages in the system will be based TLV format (Tag, Length, Value).

This document describes the characterization of managing one type of submarine, however the ground station should know how to manage several types of submarines.

> **NOTE:** The physical transport (UART / Ethernet) is a configuration detail of the communication module, not a structural assumption elsewhere in the system. All other modules interact with the communication module through a transport-independent interface.

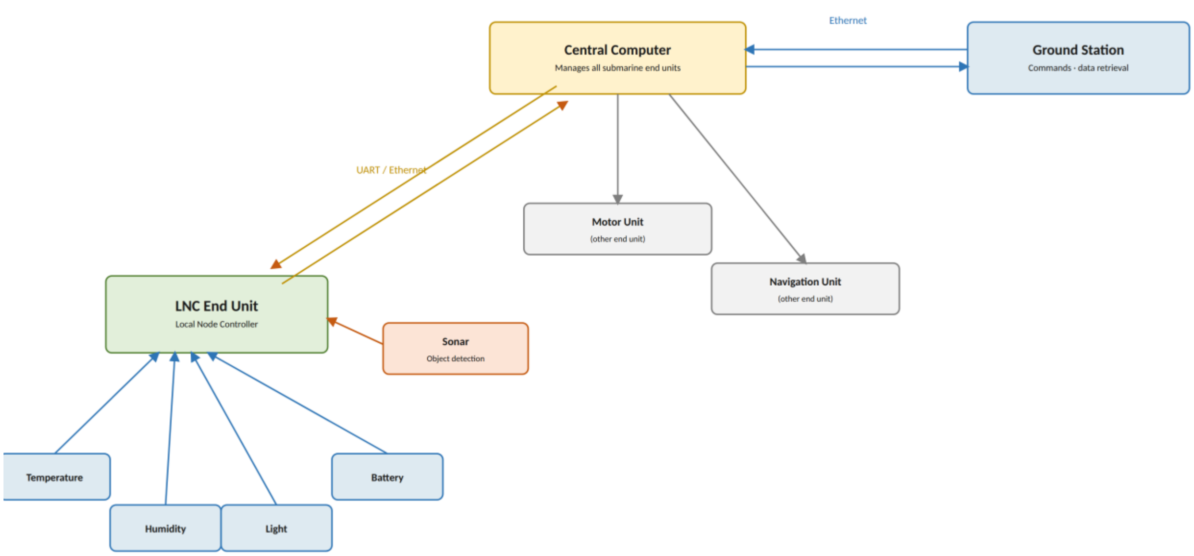

Figure 1 — System context: the LNC End Unit, its onboard sensors, and its relationship to the Central Computer and Ground Station

2. LNC End Unit

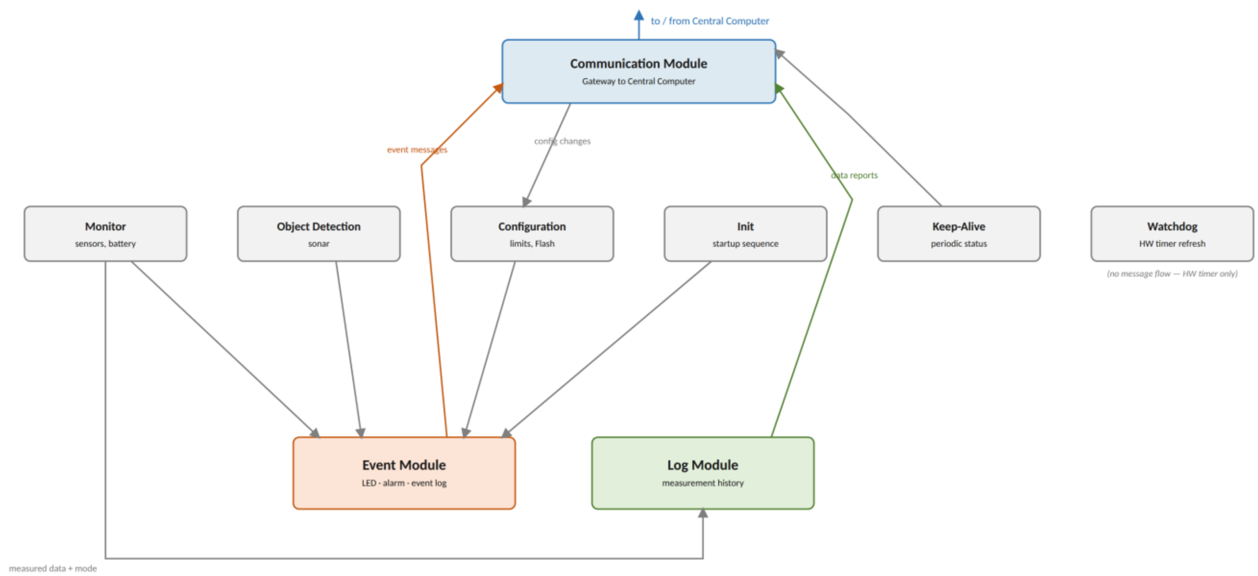

Figure 2 — LNC module interaction overview: message flow between the nine software modules

The LNC end unit is composed of the following software modules: Monitor, Object Detection, Event, Log, Communication, Configuration, Init, Keep-Alive, and Watchdog. Each module's functional behaviour is defined below, followed by the system's operating modes.

2.1 Monitor Module

- Periodically, once every 5 seconds, the module samples temperature, humidity, battery voltage (via the potentiometer), and lighting conditions, and compares each measured value against its configured limits.
- The module sends the measured data and the resulting mode (per the limit comparison) to the Log module.
- Limit values are configurable and are stored in Flash.
- If the measured data results in a mode change, the module sends a message to the Event module containing the measured values.

2.2 Object Detection Module

- Continuously listens for object detection in the submarine's direction of travel.
- Upon detecting an object, sends a message to the Event module.
- If a previously detected object is no longer detected, sends a corresponding message to the Event module.

2.3 Event Module

The Event module waits to receive events from other modules. Every event is timestamped on arrival. The module's behaviour depends on the event's source, as defined below.

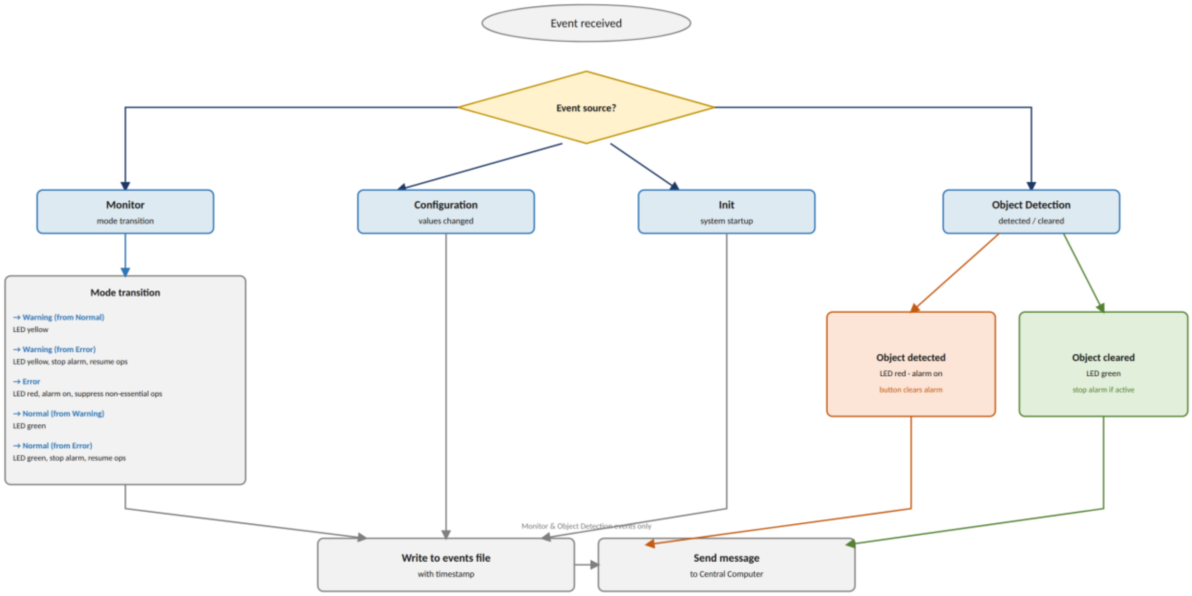

Figure 3 — Event Module decision flow: actions taken per event source and outcome

2.3.1 Events from the Monitor Module

- Transition from Normal to Warning:
  - The RGB LED is set to yellow.
- Transition from Error to Warning:
  - The RGB LED is set to yellow.
  - If the alarm sound is still active, it is stopped.
  - The system resumes full operation.
- Transition to Error:
  - The RGB LED is set to red.
  - The alarm sound is activated.
  - Pressing the button stops the alarm.
  - The system suppresses non-essential operations.
- Transition from Warning to Normal:
  - The RGB LED is set to green.
- Transition from Error to Normal:
  - The RGB LED is set to green.
  - If the alarm sound is still active, it is stopped.
  - The system resumes full operation.
- In every case above, a message including the timestamp is written to the events file, and a message is sent to the Central Computer.

2.3.2 Events from the Configuration Module

- A message including the timestamp is written to the events file.

2.3.3 Events from the Init Module

- A system startup message is written to the events file, including whether the startup followed a watchdog (WD) reset.

2.3.4 Events from the Object Detection Module

- Object detected in the direction of travel:
  - The RGB LED is set to red.
  - The alarm sound is activated.
  - A message including the timestamp is written to the events file.
  - A message is sent to the Central Computer.
  - Pressing the button stops the alarm.
- Object no longer detected in the direction of travel:
  - The RGB LED is set to green.
  - If the alarm sound is still active, it is stopped.
  - A message including the timestamp is written to the events file.
  - A message is sent to the Central Computer.

2.4 Log Module

- Builds a log message containing the timestamp, measurement data, and mode.
- Writes the message to a file named according to the date.
- Retains up to 7 days of data (7 files) at any given time. Starting on the 8th day, the file from the oldest day is deleted.

2.5 Communication Module

- Has access to the UART interface.
- Handles both sending and receiving of messages to and from the Central Computer.
- Receives Management commands from Central Computer:
  - Set the temperature range that defines the Normal mode condition.
  - Set the temperature range that defines the Warning mode condition.
  - Set the lower boundary value for humidity that defines the Normal mode condition.
  - Set the lower boundary value for humidity that defines the Warning mode condition.
  - Set the lower boundary value for lighting that defines the Normal mode condition.
  - Set the lower boundary value for lighting that defines the Warning mode condition.
  - Set the lower boundary value for battery voltage (potentiometer) that defines the Normal mode condition.
  - Set the lower boundary value for battery voltage (potentiometer) that defines the Warning mode condition.
  - Set the date and time for the RTC.
  - Get the current system time.
- Receives instructions from the Central Computer:
  - Retrieve measurement data for a given time range.
  - Retrieve events for a given time range.
- Sends messages originating from other modules, by priority:
  - Keep-alive messages — highest priority.
  - Event messages — medium priority.
  - Data reports — low priority.

2.6 Configuration Module

- Receives configuration changes from the Communication module and manages all configuration values for the system.
- Persists configuration values in Flash.
- Loads configuration values from Flash at system startup. If Flash contains no data (first boot), default values are loaded and written to Flash.

2.7 Init Module

- Requests time and date synchronization from the Central Computer, via the Communication module.
- Sends a message to the Event module once synchronization completes.
- Starts all system activities.

2.8 Keep-Alive Module

- Periodically, every 6 seconds, sends a keep-alive message to the Central Computer via the Communication module. The message includes the timestamp, the latest measurement data, and the current system mode.

2.9 Watchdog Module

- Refreshes the hardware watchdog (WD) timer on the correct schedule to prevent a watchdog-triggered reset.

2.10 Operating Modes

The LNC operates in one of three modes at any given time, determined by the Monitor module's comparison of measured values against configured limits:

Mode	Condition
Normal	All measurements are within the Normal range.
Warning	At least one measurement is within the Warning range, and no measurement is within the Error range.
Error	At least one measurement is within the Error range.

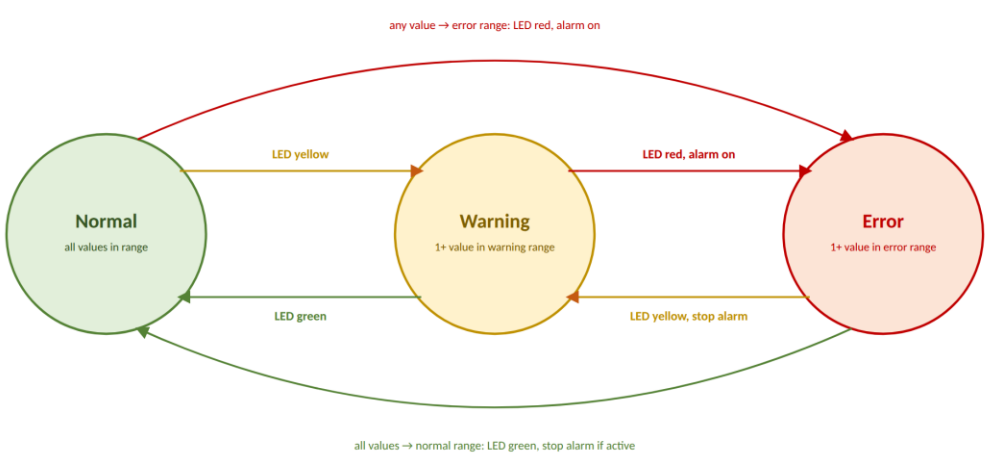

Figure 4 — Operating Mode transitions: Normal, Warning, and Error, with the LED and alarm action for each transition

3. Central Computer

The Central Computer is responsible for the following software modules: Communication (with the LNC end unit), Management Command, Log, and Data Collection & Analysis.

He manages all the LNCs, gives them orders and receives information from them. He processes all the information and makes decisions about how to continue the mission.

According to LNC, it also only stores one week's worth of information (last seven days).

3.1 Communication Module (LNC-Facing)

- Implements the protocol for sending and receiving messages to and from the LNC end unit.
- Implements a listener that forwards received messages to the appropriate modules.
- Communicates with the LNC end unit over UART or Ethernet; the module is built so that the transport can be changed to another technology with minimal impact elsewhere in the system.

3.2 Management Command Module

- Implements Management commands to be sent to the LNC end unit via the Communication module.
- Handles command processing.

3.3 Log Module

- Prints logs and persists them to files.

3.4 Data Collection & Analysis Module

- Persists data received from the LNC device in the database.
- Receives events and persists them in the database.
- Prepares reports from the stored data, broken down by various criteria.

4. Ground Station

The ground station can request from the submarine's central computer data stored over a period of time (date and time), both the data stored in a log and data of the events that occurred over the period of time.

OOP Part

Submarine Fleet Management System
In this project, we will manage a system for a submarine fleet. The system will store information about the different submarines in the fleet, the missions assigned to them, and the communication between submarines during a mission.
Each submarine has a **serial number** and a **name**. In addition, for every submarine the system must indicate whether it is **currently assigned to a mission** or available for a new mission.
The fleet contains two types of submarines: **research submarines** and **combat submarines**.
A research submarine is used for research missions. For each research submarine, the system must store the **names of the researchers on board** and the **current research topic**.
A combat submarine is used for operational missions. For each combat submarine, the system must store the **description of its current mission**, the **name of the commander**, and the **number of combat personnel on board for the mission**.

Each submarine has a **central computer** that works with the submarine's sensors and receives information from them. The central computer itself is defined in the first part of this document, but it should be treated as an object that belongs to each combat submarine.

The system must support adding submarines of the different types to the fleet, assigning missions to submarines, and updating the relevant mission data. When a submarine is assigned to a mission, its status must also be updated so that it is clear that it is no longer available for another mission.

During a mission, a combat submarine may operate together with additional combat submarines. Therefore, for each combat submarine, the system must also keep track of the other submarines participating with it in the current mission, and the all the missions that its fleet was managing.

Submarines participating in the same mission can communicate with one another and send messages. Whenever a submarine receives a message, the system must store the **message content** together with a **reference to the submarine that sent it**, so that the sender of each message can be identified.

Operations Supported by the Menu

The system menu should allow the user to perform the following operations:

1. **Add a new submarine to the fleet** - choose the submarine type and enter the relevant details.
2. **Display all submarines in the fleet** - including the relevant details of each submarine and whether it is currently assigned to a mission.
3. **Search for and display a submarine by serial number**.
4. **Assign a mission to a submarine** - enter the mission details and update the submarine as assigned to a mission.
5. **Update a submarine's mission details** - according to the submarine type.
6. **End a submarine's mission** - remove the mission assignment and mark the submarine as available for a new mission.
7. **For combat submarine - associate additional combat submarines with the same mission**.
8. **Send a message from one combat submarine to another submarine participating in the same mission**.
9. **Display the messages received by a submarine**, including the content of each message and the submarine from which it was sent.
10. **Exit the system**.